Annex E

FARTMAPT.M Program Code (*Matlab 4.0*)

```
function fartmapt(FormFile,MODE,SIV,PlotFlag)

%          A FUZZY ART MAP NEURAL NETWORK
%               SPEECH RECOGNITION SYSTEM
%                         BASED ON
%                    FORMANT RATIOS
%
%                          by
%         Capt D.P. Young, CD, B.Sc., RRMC
%                   Canadian Forces
%
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% This program may be used within the Department of National Defence
% but copyright for open use remains the property of the author.
%
%
% Usage fartmapt(FormFile,MODE,SIV,PlotFlag)
%
%   FormFile = string name of file containing extracted formant vectors
%MODE = 1 for training
%          = 2 for testing
%     SIV = Signal Identification Vector (used to map fuzzy ART categories to a given
phrase)
%           = 999 for autodetect from filename
% PlotFlag = 0 for no graphics
```

```
%            = 1 for graphics

% TURN ON WHITE BACKGROUND FOR SCREEN SHOTS
% ensure all figures are closed first
close; close; close; close; close; close; close; close; close; close; close;
whitebg;

%~~~~~~~~~~~~~~~~~~~~~~~~~ GLOBAL CONSTANTS ~~~~~~~~~~~~~~~~~~~~~~~~~
if MODE==1
  vigilance=1.00; % This is the baseline training vigilance
else
  vigilance=0.50; % This is the baseline testing vigilance
end;
% The choice function uses this non-zero positive parameter in its denominator to
% break the tie between competitive nodes
Choice_Parameter = 0.001;


NewClusterFlag = 0; % This flag is used to signal the creation of a new cluster

% Code number used to label unrecognized input vectors. A positive number is used
UNRECOGNIZED=999;
 % since it must serve later as an vector index (no negative indexes allowed)
 % Warning: Ensure this number is not overrun by a real cluster number due to
 % cluster expansion(explosion?).

StandardSIVLength=80; % Max length of SIV string, used to pad with blanks f
%~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
% BATCH MODE SIV ASSIGNMENT BY FILE NAME
%~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
```

```
% The SIV arguement is set to 999 when the program is to automatically construct a SIV
code
% based on the file name of the formant file.

if SIV(1) == 999
      BatchFlag=1;
         PHRASE=FormFile(8);
         if PHRASE=='0' SIV='DATE AND TIME';
         elseif PHRASE=='1' SIV='NAME';
         elseif PHRASE=='2' SIV='SHE HAD YOUR DARK SUIT IN GREASY WASH
WATER ALL YEAR';
         elseif PHRASE=='3' SIV='DON^T MAKE ME CARRY AN OILY RAG LIKE
THAT';
         elseif PHRASE=='4' SIV='MAY WE ALL LEARN A YELLOW LION ROAR';
         elseif PHRASE=='5' SIV='WE WERE AWAY A YEAR AGO';
         elseif PHRASE=='6' SIV='I KNOW WHEN MY LAWYER IS DUE';
         elseif PHRASE=='7' SIV='EVERY SALT BREEZE COMES FROM THE SEA';
         elseif PHRASE=='8' SIV='OUR YACHT SLID AROUND THE POINT INTO THE
BAY';
         elseif PHRASE=='9' SIV='I WAS STUNNED BY THE BEAUTY OF THE VIEW';
         else SIV='BATCH FILE NAME ERROR';
         end; %if PHRASE
  else
      BatchFlag=0;
end; %if SIV

SIV=[SIV,blanks(StandardSIVLength-length(SIV))];
% pad SIV string to standard SIV length with blanks

FileNameSIV=SIV;
% store phrase identifier from filename for automatic error detection using database files
```

```
%#########################################################################
% PERFORM FEATURE VECTOR REPRESENTATION PRIOR TO NETWORK INPUT
%#########################################################################
% Load formant vectors from FormFile

load(FormFile);

% Calculate the Tri Formant Ratio Vector (TFRV) representation

F3F2=(Formant(3,:)./Formant(2,:));
F2F1=(Formant(2,:)./Formant(1,:));
F3F1=(Formant(3,:)./Formant(1,:));
TFRV=[F3F2,F2F1,F3F1];

% Convert the TFRV to Normalized Polar Angle (NPA) form

NPA_TFRV=atan(TFRV)/(0.5*pi);

% Linear Time Warp (LTW) the Normalized TFRV to standard network input length

StandardLength=500;
LengthTFRV=length(TFRV);

for Index=1:StandardLength-1
   LTW_NPA_TFRV(Index)=NPA_TFRV(ceil(LengthTFRV/StandardLength*Index));
end;

LTW_NPA_TFRV(StandardLength)=NPA_TFRV(LengthTFRV);
% this is a work around a bug in the ceil function
```

```
if PlotFlag > 0
        % Plot the feature vector
        figure(1);
        subplot(3,1,1);
        plot(LTW_NPA_TFRV,'k');
        axis([0 StandardLength 0 1]); % a standard axis aids in visual comparison
        title(['Tri Formant Ratio Vector from ',WaveFile]);
        ylabel('NPA Ratio');
        grid on;
end;

% Initialize starting weights:
% Load Network Weight Matrix file if it exists or initialize from scratch
% NOTE: floppy drives can be used for removeable media

FID = fopen('fartmapt.mat'); % (FID = -1 if it doesn't exist)
if FID == -1
   fprintf(' \n INITIALIZING Fuzzy ART Map Network Memory Matrix ! \n');
   M=2*StandardLength;  % The input dimension of the network
% It is TWICE THE SIZE OF THE FEATURE VECTOR due to complement coding
   N=10;
% N is the number of outputs from the network.
% It defines the number of classifications defineable - matrix is initialized with 10 outputs
% MORE OUTPUTS ARE ADDED AS DEMANDED BY PROGRAM NEEDS
   W = ones(N,M);
% W is the feedforward/feedback weight matrix between the input and output layers
   max_cluster_number = 0;
% keeps track of the highest committed cluster
   SIM=ones(N,length(SIV));
% Signal Identification Matrix with length(SIV) outputs and N category inputs
```

```
% NOTE** This convention is currently the reverse of that used in W weight matrix
% Recode the W matix to (input#,output#) standard at a later date (reduce code)
else
   fclose(FID);
   load('fartmapt.mat');
   [N,M]=size(W);
end;

%#########################################################################
% FUZZY ART SELF ORGANIZATION
%#########################################################################

% Calculate complement coding input vector..Note that the dimension has doubled.
I = [LTW_NPA_TFRV,1-LTW_NPA_TFRV];
% it is assumed that the feature vector is a row vector

% Calculate choice function Tj(I) for each output node
% To save time, PERFORM CHOICE FUNCTION ONLY FOR CATEGORY NODES
% AND ONE UNCOMMITTED NODE
for j = 1:(max_cluster_number+1)
   T(j) =  sum( min(W(j,:),I) ) / ( Choice_Parameter + sum(W(j,:)) );
end; % for j

% Perform "Hypothesis" testing, in descending order of Choice Function output magnitude
% until a match is made that meets the required vigilance, and meets the Match Tracking
% requirements

active_nodes = (max_cluster_number+1);
% identify number of nodes in competition (other outputs are spares)

MatchTracking=1; % set flag
```

```
while MatchTracking==1

 cluster_search = 1; % turn flag on
 while cluster_search == 1,

   % find node with maximum output
   % IF TWO NODES BOTH HAVE MAX OUTPUTS THEN THE NODE WITH THE
   % LOWEST INDEX GETS CHOSEN , max function does this automatically
   [maxnum,max_index] = max(T);

   % Vigilance (similarity) test

   similarity = sum( min(W(max_index,:),I) ) / sum(I);

   if (similarity < vigilance) & (active_nodes > 1)

      % RESET: Disable Node and look for next T(j)max

      T(max_index) = 0; % deactivate winning node by setting its choice function to zero
      active_nodes = active_nodes - 1; % decrement active node counter

    else % RESONANCE: stop search and record cluster number

        cluster=max_index;
      cluster_search = 0; % set flag to end cluster search, exit from while loop

  end; % if similarity
end; % while cluster_search

if MODE==1
    % >>>>>>>>> LEARNING MODE - MATCH TRACKING PROCESS <<<<<<<<
```

```
    % if the cluster output achieved by the resonance (above) is not mapped to the desired
    % SIV code in the Signal Identification Matrix (SIM) then the vigilance parameter is
    % increase from the baseline value to a value just above the similarity ratio (in order to
    % induce reset) and the "Hypothesis" testing is restarted from where it left off

    if SIM(cluster,:)==SIV    % Check for SIV match and break out of while loop
        MatchTracking=0;
    elseif max_index > max_cluster_number  % Check for new cluster, and assign SIV to
SIM
      SIM(cluster,:)=SIV;
      MatchTracking=0; % Break out of while loop
    else
        % Boost vigilance just above similarity
        vigilance = 1.00000000000001*similarity;
       % Matlab displays accuracy to 14 decimal places
    end; %if SIM
  else
    MatchTracking=0; % Only perform Match Tracking in learning mode
  end; % if MODE==1

end; % While MatchTracking==1


% LEARNING PHASE:     FAST LEARNING = NO RECODING OF CATEGORY
% WEIGHTS AFTER INITIAL FORMATION ... A new category is formed solely around a
% new input vector which did not fit in any other category

if max_index > max_cluster_number  % current max_index is a new cluster
   if MODE ==1
        % Update flags, counters and weights
        NewClusterFlag = 1;
```

```
        max_cluster_number = max_index;
        fprintf('\nNew Cluster Formed = %i \n',max_cluster_number);
      W(max_index,:) = I; % Because the learning rate is 1 all new clusters are centered on the founding vector
    else
      % MODE 2 RECOGNITION ONLY - NO LEARNING ALLOWED
      fprintf('\n UNRECOGNIZED FEATURE !!! \n');
      cluster=UNRECOGNIZED; % assign the feature vector to the "UNRECOGNIZED" cluster instead of creating a new cluster
    end; % if MODE
end; % if max_index

% If in test mode, then Identify the voice sample
if max_index <= max_cluster_number
  if MODE==2
      SIV=SIM(cluster,:);

% AUTO ERROR DETECTION FOR DATBASE FILES -
% compare SIV of matching pattern to that of input file name
      if (BatchFlag == 1) & (strcmp(SIV,FileNameSIV)==0)
            fprintf('WRONG PHRASE IDENTIFIED || ');
        end;

        % PRINT OUT MATCHING VECTOR INFO
      fprintf('%5.2f Percent || %s \n',100*similarity,SIV);

        % PLOT THE MATCHING MEMORY VECTOR FOR COMPARISON
        if PlotFlag > 0
              figure(1);
              subplot(3,1,2);
              plot(W(cluster,:),'k');
```

```
            axis([0 StandardLength 0 1]); % a standard axis aids in visual comparison
            title('Fuzzy ART Map Stimulus Response');
            ylabel('NPA Ratio');
            grid on;


            % OVERLAP THE TWO VECTORS
            subplot(3,1,3)
            plot(LTW_NPA_TFRV,'k')
            axis([0 StandardLength 0 1]); % a standard axis aids in visual comparison
            title([num2str(100*similarity),'% Match to ',deblank(SIV)]);
            xlabel('Linearly Warped Time Scale')
            ylabel('NPA Ratio');
            hold on
            plot(W(cluster,:),'k');
            hold off
            grid on;
        end;

   end; % if MODE==2
end; % if max_index

%#########################################################################
% NEW IDEA -
% allow weight matrices to grow when it gets low on uncommitted output classes
% ADD WEIGHTS FOR TWENTY MORE CATEGORIES
  if (N - max_cluster_number) < 2
     N=N+20 ;
     fprintf('\n >>>>>>>>>>> Caution - Network Memory is GROWING!!! <<<<<<<<');
     W=[W;ones(20,M)];
     % Expand Signal Identification Matrix to accomodate 20 more input categories
```

```
    SIM=[SIM;ones(20,length(SIV))];
  end;
%
%#########################################################################

% PLOT FINAL WEIGHTS IF THEY HAVE CHANGED DUE TO NEW
% CLUSTER FORMATION
if NewClusterFlag == 1 & PlotFlag > 0
   figure(2);
   imagesc(W');
   axis('xy');
   title('Fuzzy ART Map Memory');
   xlabel('Long Term Memory (LTM) Patterns');
end;

% SAVE NETWORK WEIGHTS AND NUMBER OF LAST COMMITTED OUTPUT
% NODE
if NewClusterFlag==1
        save fartmapt.mat W max_cluster_number SIM
end;

% PRINT END OF PROGRAM
fprintf('=========================================================\n');
```